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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

**ELECTRONIC CITY CAMPUS,
BENGALURU – 560100, KARNATAKA, INDIA**

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IV SEMESTER**

**UE23CS251B – MICROPROCESSOR AND COMPUTER
ARCHITECTURE LABORATORY**

PROJECT REPORT

ON

Smart Glove-Based Sign Language Detection System

SUBMITTED BY

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ABSTRACT OF THE PROJECT:

In this project, we present a smart glove system capable of recognizing sign language gestures using four flex sensors and an MPU6050 motion sensor. The glove captures finger bending (via flex sensors) and hand orientation/movement (via the MPU6050), and transmits the data wirelessly using an HC-05 Bluetooth module. These sensor readings are processed to detect specific hand gestures, which are then converted into corresponding text outputs. The system provides a low-cost, real-time solution to bridge communication gaps between speech-impaired and hearing individuals, emphasizing portability, accuracy, and ease of use.

The microcontroller reads the analog values from the flex sensors and digital data from the MPU6050 through I2C communication. Each gesture is defined by a unique combination of finger bend and hand position, which is matched with predefined patterns in the software. The HC-05 module allows seamless wireless communication with external devices such as smartphones or computers, where the translated text can be displayed, enhancing user interaction.

This glove can be expanded to include additional sensors for more complex gestures or even integrate with voice-based systems for multi-modal communication. Future iterations may involve machine learning algorithms for dynamic gesture training, improving adaptability to different users' signing styles. The modular and open-source nature of the project makes it ideal for educational purposes, assistive technology development, and hobbyist innovation.

CIRCUIT DIAGRAM:

